

6-year statistical analysis of cloud occurrence and its physical properties using ACTRIS-CCRES remote sensing data

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1 Introduction

Clouds play a vital role in regulating Earth's radiative budget and hydrological cycle. They interact with solar and thermal radiation, affecting the energy balance and, consequently, the surface temperature (Twomey, 1977; Brenguier et al., 2003; Zeng et al., 2021). Additionally, clouds transport moisture to various regions, being directly associated with drought, which is intensifying in the Mediterranean Region (Rios-Entenza et al., 2014; Hoerling et al., 2012). However, studying clouds is quite complex, and their radiative effects are in the spotlight of the atmospheric science community. This challenge arises from clouds' high spatial and temporal variability, as well as the variety of hydrometeor types inside them (int, 2023). The hydrometeor phase, shape and concentration can significantly alter their interaction with solar and thermal radiation (Yoshida and Asano, 2005), posing a significant challenge to estimate their radiative effect as suggested by ?.

The ground-based remote sensing technique is one of the state-of-the-art methods for measuring clouds. Particularly, the Cloud Doppler Radar (CDR) can perform long-term measurements of clouds, which is one alternative for improving the cloud knowledge. Several studies have been analyzing CDR data in Europe, but they are mostly located in Northern Europe (Kneifel et al., 2022; Nomokonova et al., 2019; Bühl et al., 2016; Achtert et al., 2020). In this context, the AGORA (Andalusian Global ObseRvatory of the Atmosphere) ACTRIS CCRES station has been operating a 94 GHz CDR since 2018, contributing to enhancing the lack of data on the Mediterranean basin, which is a key region for climate variability. This station is one of the southernmost stations that is part of the Cloudnet network (Illingworth et al., 2007). Notably, it stands as the only cloud remote sensing database available in Spain, making it an invaluable resource for regional climate studies. Satellite data would be an alternative to fill the gap of data, but they only provide a general picture of cloud properties. They present low temporal resolution and information on low level clouds is affected by large uncertainties. Additionally, this data should be validated with the use of ground-based remote sensors such as lidars and CDRs (Protat et al., 2006, 2010). Thus, ground-based observations are still necessary for improving the accuracy of cloud properties.

This study aims to perform a statistical analysis on single layer clouds for more than half a decade of observations at the AGORA station. First, clouds were classified (e.g. ice, liquid or mixed-phase clouds) using the cluster classification algorithm (CBA). This is a novel algorithm that classifies clouds according to the percentage of hydrometeors within each cluster. The clusters are defined using the hydrometeor classification product from Cloudnet. Second, cloud macrophysics (e.g., cloud thickness, base and top height) and microphysics statistics are performed for each cloud category from CBA. Markedly, cloud statistics in this study are derived from a cluster-based approach, contrasting with previous studies that used the profile-based algorithm (PBA) (Pîrloagă et al., 2022; Nomokonova et al., 2019; Kneifel et al., 2022). For instance, Nomokonova et al. (2019) used the PBA algorithm to characterize single-layer clouds at the Ny-Ålesund station, Kneifel et al. (2022) employed it to investigate ice cloud variability in the German Alps, and Pîrloagă et al. (2022) applied it to examine seasonal cloud variability in Bucharest. Bühl et al. (2016) also uses the PBA but selected intervals with more than 15 min with the same cloud category. On the other hand, the CBA considers the composition of the entire volume occupied by the cloud. Roschke et al. (2024) also used the CBA, but with a different classification scheme, separating clusters into warm and cold clouds for the Barbados site.

The experimental site, datasets, instruments and products are described in Section 2. The proposed new approach on cloud classification algorithm and the cloud assessment are presented in Section 3. Section 4 analyzes the annual, seasonal, and daily cycle of cloud types, as well as the base and top height, and thickness. In addition, this section discusses the liquid and ice properties variability for different clouds. Finally, Section 5 summarizes the main findings and discusses their importance to improve further cloud statistics analysis.

2 Experimental site and Instrumentation

The AGORA ACTRIS-CCRES station (37.16°N, 3.61°W, 680 m a.s.l) is located in the South of the Iberian Peninsula being one the most meridional CLOUDNET stations. This station has been part of Cloudnet since 2018 and was integrated into ACTRIS CCRES (where ACTRIS stands for Aerosols, Clouds, and Trace gasses Research InfraStructure, and CCRES for Centre for Cloud Remote Sensing) in 2023 as a new national facility. The region experiences a continental climate, with colder months in winter and spring, and warmer months in summer and fall. Particularly, between summer and winter, the temperature and humidity presented the largest differences (Bedoya-Velásquez et al., 2019). The relative humidity below 5 km a.g.l ranges between 60-75% during winter, and 40-50% during summer. These thermodynamic conditions will be particularly important to understand diurnal cycle of cloud types and inter-annual differences in cloud properties analyzed in further sections.

2.0.1 Instrumentation

In this study we use a synergistic combination of a MicroWave Radiometer (MWR), a ceilometer and a 94 GHz Cloud Doppler Radar (CDR). These instruments have been widely used for many Cloudnet stations and shows a promising configuration for long-term cloud observations (Illingworth et al., 2007), especially to calculate cloud microphysics with better accuracy than satellites (Protat et al., 2009). These instruments are operated in AGORA and their relevant technical specifications are shown

in table 1. This table details operational frequencies or wavelengths, type of measurements, products, measurement ranges,
 55 data collection intervals, and references for more details.

The RPG-HATPRO G2 (Humidity and Temperature PROfiler) MWR is a passive remote sensing instrument that measures the brightness temperature (T_B) around the water vapor (22-31 GHz) and oxygen (51-58 GHz) absorption bands. The instrument is used to retrieve liquid water path (LWP) and integrated water vapor (IWV) (see Tab. 1). Their principal characteristics are radiometric accuracy of 0.3-0.4 K and integration time of 1 s. More details can be found in Navas-Guzmán et al. (2014);
 60 Rose et al. (2005).

The CHM15k Nimbus ceilometer is an active remote sensing instrument that utilizes a vertically pointed Nd:YAG pulsed laser to measure backscattered photons. The instrument is used to retrieve the attenuated backscattering coefficient (β) of aerosols and small cloud droplets. Their principal characteristics are repetition frequency intervals of 5-7 kHz, where each pulse is emitted at 1064 nm with 8.4 μ J of energy. The laser beam divergence is 0.3 mrad, and range and time resolution of
 65 15 m and 15 s, respectively. Also, the overlap completeness at 1500m a.g.l (Heese et al., 2010). More details can be found in (Cazorla et al., 2017).

The dual polarimetric RPG-FMCW 94 GHz CDR is also an active remote sensing instrument that measures the Doppler Velocity Spectrum (DVS) for horizontal and vertical linear polarizations at 94 GHz. The instrument is used to compute the reflectivity (Z), mean Doppler velocity (ν_D), and spectral width (S_w). The CDR also derives the linear depolarization ratio
 70 (LDR), enabling the detection of the melting layer, aerosols, and insects as described by Hogan and O'Connor (2004). Their principal characteristics are the frequency-modulated continuous wave (FMCW) signal, which allows different range and time configurations (see Tab. 1), the frequency range of 300-3700 kHz and, the antenna beam width of 0.56 $^\circ$.

Table 1. Instrument specification and data products

Instrument	Bands / wavelengths	Measured	Pos-processed	Range	Time	References
94-GHz Cloud Doppler Radar	94 GHz (W-band)	DVS	ν_D , Z , δ , S_w , LDR	12.8 to 51.1 m ^a	1.1 to 3.6 s ^b	Küchler et al. (2017)
RPG HATPRO	22-31 GHz (K-band) 51-58 GHz (V-band)	T_B	LWP	-	1 s	Navas-Guzmán et al. (2014)
CHM15k Nimbus Ceilometer	1064 nm		β ($\text{sr}^{-1} \text{ m}^{-1}$)	15 m	15 s	Cazorla et al. (2017)

^a This range varies with the height and depends on the chirp table configuration.

^b It also depends on the chirp type.

The Cloudnet processing chain (Illingworth et al., 2007) performs extensive quality assurance on raw instruments for providing high-level products, such as target classification, liquid water path, and droplet effective radius. This chain standardizes
 75 data processing within the cloud remote sensing community. Their products are derived using a synergistic approach, integrating vertically pointing measurements from ground-based CDRs, MWRs, and ceilometers. Specifically, the target classification

product, generated through these three instruments, has been utilized in numerous studies for cloud typing and multi-year analysis (Pîrloagă et al., 2022; Nomokonova et al., 2019; Kneifel et al., 2022), as already commented.

We used a large dataset covering the years 2018 to 2023. Figure 1 illustrates the monthly Cloudnet high level products availability for the AGORA ACTRIS-CCRES station. The availability of this data depends on simultaneous vertically pointing measurements of the RPG-HATPRO G2, CHM15k Nimbus ceilometer, RPG-FMCW 94 GHz CDR, as well as meteorological model products from the ECMWF (European Centre for Medium-Range Weather Forecasts) at our station. As can be seen in Figure 1, the months that less meet these conditions are Jan-Mar, with 40-50% of the data available, whereas from Apr to Oct the data availability is over 60%. The years 2018 (270,000 profiles) and 2020 (968,769 profiles) have the least and largest amount of data, respectively. The white bars denote periods with missing data due to instrument maintenance, technical issues, and scanning measurements (which are not processed by Cloudnet). It is worthy to note that most of the missing data are days without clouds, as will be clear in section 4.1. Thus, the recorded dataset shows a solid database, with enough samples to perform an statistical analysis.

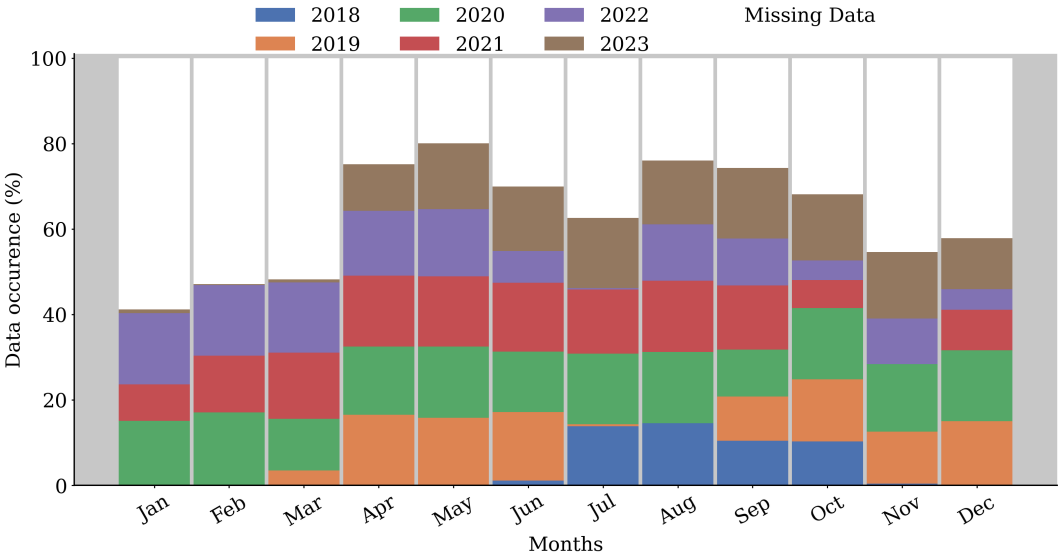


Figure 1. High level post-processed data availability per month at the AGORA ACTRIS CCRES station. This dataset represents more than half a decade of synergic ground-based measurements at this station. Each color represents the data availability for different years.

2.0.2 Cloudnet products

The relevant Cloudnet products in this study include only target classification (Hogan and O'Connor, 2004) and microphysical retrievals, such as effective radius, and liquid and ice water content. A resumed description of microphysical products assumptions and uncertainties are shown in table 2. These products are re-gridded to time resolution of 30 s and same radar range resolution. Nevertheless, these products can be highly affected by uncorrected radar reflectivity values. For high-frequency

radars, such as the RPG models, this variable suffers a strong attenuation by liquid water and gases (Hogan et al., 2003).

95 Hence, Cloudnet applies the two-way gas and liquid attenuation correction for the reflectivity.

The target classification product (TC) assigns an integer (0-10) to atmospheric particles by height and time. The particle names and its respective numbers are the following: Clear sky (0), Droplets (1), Drizzle or rain (2), Drizzle & droplets (3), Ice (4), Ice & droplets (5), Melting ice (6), Melting & droplets (7), Aerosol (8), Insects (9), Aerosol & insect (10). Nevertheless, we are interested only in hydrometeors, thus we will refer to HC (hydrometeor classification) as only pixels classified as
100 hydrometeors i.e. from 1 to 7. Briefly, these pixels are classified as follow: Liquid and supercooled droplets were identified using lidar attenuated backscatter and CDR Doppler velocity spectra (Schimmel et al., 2022), applied only at temperatures above -40°C. Rain was identified as falling liquid particles below the cloud base, while ice was attributed to particles with wet-bulb temperatures below freezing. The melting layer was detected using wet-bulb temperature divergence and LDR from CDR.

105 For the AGORA station, many cases of hydrometeor misclassification on HC were observed. Figure 2.a shows a thin and lower precipitable ice cloud between 08:00 and 20:00. However, at this time, there are just insects and aerosols in the atmosphere, as can be checked by small values of attenuated backscattering (Fig. 2.b) and CDR reflectivity

remove cloud classification and radar reflectivity

(Fig. 2.c). This occurs because insect detection is highly dependent on the model temperature and the LDR (Roschke et al.,
110 2024). Due to the absence of LDR in many stored CDR data files at our station, insect detection is suboptimal in these cases. However, these cases were easily identified and filtered as follow: Precipitating and non precipitating ice clouds which were identified with mean cloud base below 4 km (above mean sea level) and average cloud thickness below 700 meters were reclassified as noise. These thresholds were empirically determined by analyzing many different days of target classification products at the Granada station.

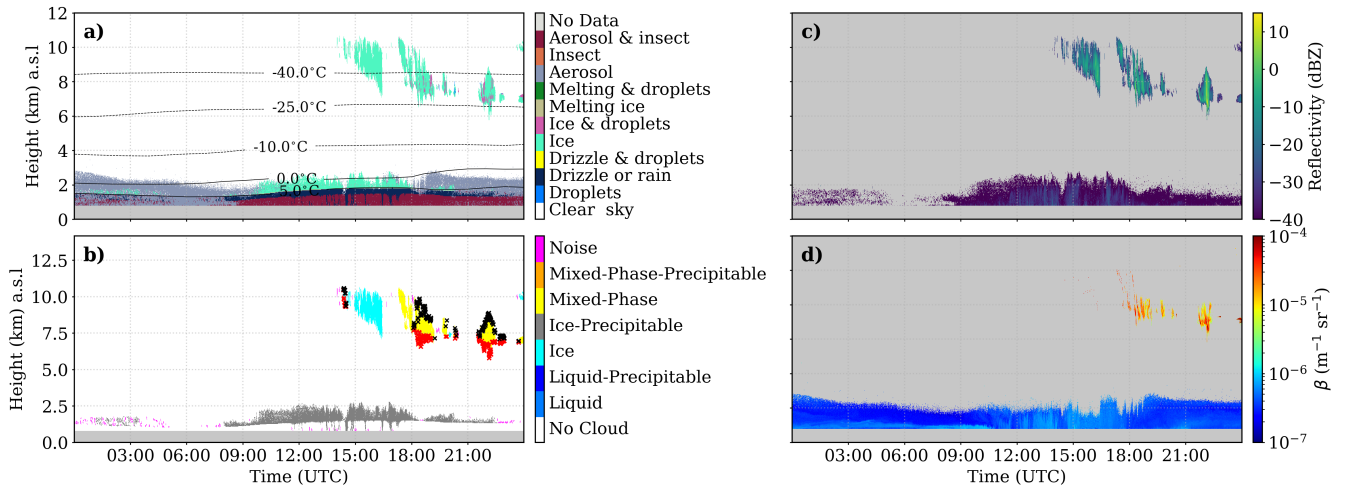


Figure 2. A case study of Cloudnet hydrometeor misclassification; (a) Target classification product; (b) Attenuated backscattering coefficient from ceilometer, which shows no assignatures of clouds in the ABL

115 Microphysical properties are retrieved in hydrometeor pixels classified by HC. Liquid water content (LWC) and droplet effective radius (r_{liq}) are retrieved at “Droplets”, “Drizzle & droplets” and “Ice & droplets” pixels. Whereas, Ice water content (IWC) and ice effective radius (r_{ice}) are retrieved only in Ice pixels. Table 2 shows inputs and output relative error for each microphysical propertie. For LWC, LWP from MWR, temperature (T_M) and pressure (P_M) from ECMWF model were used as input parameters. The output has 15% of relative uncertainty (Frisch et al., 1998). For r_{liq} , reflectivity (Z) from CDR and default values of lognormal droplet size distribution (DSD) were used as input. The output has 15% of uncertainty due to Z errors (Frisch et al., 2002). IWC is retrieved troughout the Z-IWC-T relation (Hogan et al., 2006), where Z and T_M were used as inputs. The output has a relative error ranging from +55% to -35% for temperatures between -20°C and -10°C, and +90% to -47% for temperatures below -40°C. Particularly, its main source of error is Z, where 3% of error in Z can lead to a 12% of error in ice water path (IWP), as observed by Saavedra Garfias et al. (2023). Lastly, r_{ice} is retrieved troughout the Z-T relation (Griesche et al., 2019), where Z and T_M were used as inputs. The output has 50% of relative uncertainty. These uncertainties are quite high, highlighting the challenges of accurately retrieving ice properties.

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3 Methodology

3.1 Cluster-based algorithm and cloud structure computation

The cluster-based algorithm (CBA) is a novel algorithm for cloud classification. The CBA classifies clusters of hydrometeors based on their composition i.e the percentage of different hydrometeor inside the cluster. The algorithm also classifies clouds as single-layer or multi-layer based on time. Figure 3 illustrate algorithm steps (numbered 1 to 7), which are described as follows:

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1. Cloud classification

Table 2. Cloudnet microphysics products used in this study, their inputs and assumptions

Cloudnet microphysics	Input specification	Relative standard deviation	Reference
Liquid water content (LWC)	T_M and P_M	15%	Frisch et al. (1998)
Ice water content (IWC)	Z and T_M	55% -35% T1	Hogan et al. (2006)
Droplet effective radius (r_{liq})	Z and lognormal parameters (N and σ_x)	15%	Frisch et al. (2002)
Ice effective radius (r_{ice})	Z and T_M	50%	Griesche et al. (2019)

(a) Cloud Mask: A cloud mask is generated from the CLOUDNET target classification

(b) Cluster Identification: Cloud pixels separated only by two pixels in any direction (analogous to 30 s - 1 min, and 20 - 50 m of height) are considered part of the same cloud. This is reached using a pixel dilation and contraction technique (Dougherty, 2018), expanding and contracting each cluster (see Appendix B for a more detailed explanation of the method). In other words, clusters that are very close to each other will be merged

(c) Cloud Criteria: Whether the cluster has more than 100 pixels, it is considered a cloud (without considering Drizzle or rain pixels). Otherwise, it is deemed as noise and is excluded from further analysis. This number of pixels was selected to focus on clouds of significant size.

(d) Cloud Phase: Clouds are classified as liquid, ice, mixed-phase, precipitating liquid, precipitating ice or precipitating mixed-phase. The criteria of each cloud depends on the percentage of hydrometeors, and are defined as follow:

i. Liquid criteria: The percentage of Droplets and drizzle plus the percentage of Droplets must be greater than 70%, i.e. $P(\text{Droplets and drizzle}) + P(\text{Droplets}) > 70\%$

ii. Ice criteria: The percentage of Ice must be greater than 90%, i.e. $P(\text{Ice}) > 90\%$ or the percentage of Droplets plus Ice & droplets must be less than 10%, i.e. $P(\text{Droplets}) + P(\text{Ice \& droplets}) < 10\%$. These thresholds were defined empirically after an extensive verification of case studies. We observed that thunderstorm clouds are practically made of Ice and Ice & droplets in most cases. However, they also have melting layer pixels, preventing the ice percentage from reaching 90%. That's why we also verify if there is a small percentage of water inside these clouds. Otherwise, cloud phase is classified as mixed-phase.

(e) Rain criteria: If the cloud has more than 10 pixels of Drizzle or rain, it is considered as a precipitating cloud.

2. Multi-layer classification

(a) Drizzle Removal: For precipitating Ice clouds and precipitating Mixed-phase clouds, Drizzle pixels are removed below the melting layer. Below this layer, they are considered as rain pixels, thus they are removed to avoid using these pixels for cloud base height (CBH) estimations.

- (b) Cloud Layer Criteria: For cloudy periods, time intervals where clouds overlap are categorized as multi-layer. Otherwise, clouds are categorized as a single-layer type. The minimum vertical distance between clouds to be considered as multilayer is one pixel. It was settled to avoid any possibility of considering multi-layers in further analysis.

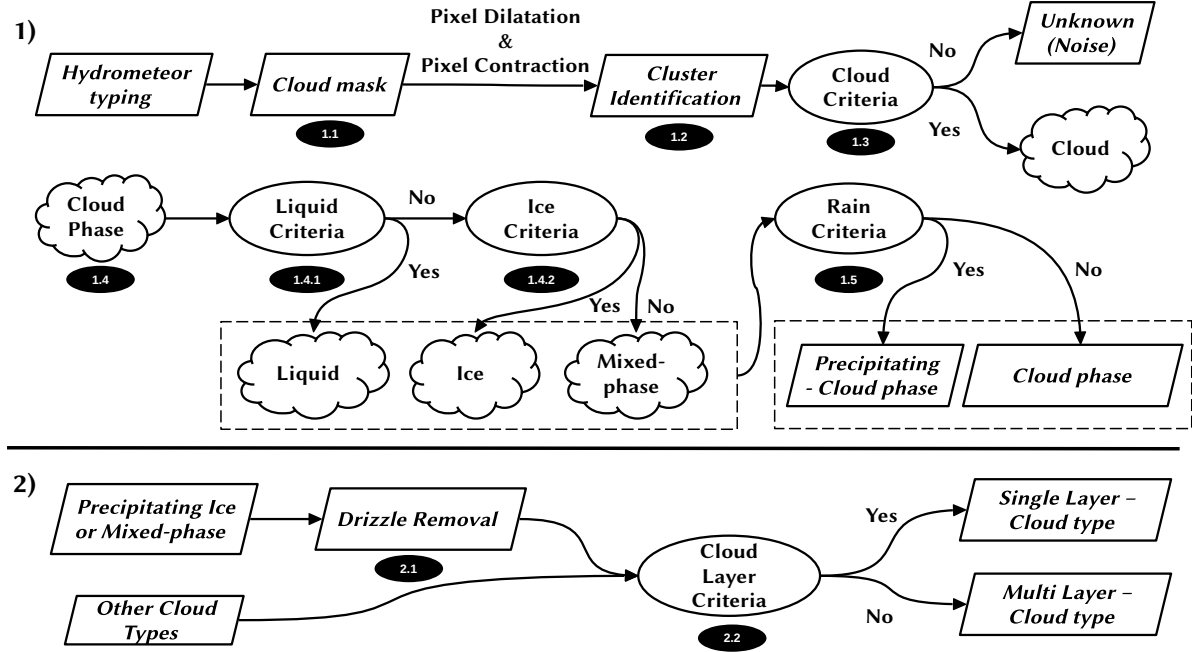


Figure 3. Flowchart of the cluster-based algorithm (CBA). Details of each step are provided in the text.

For single layer clouds, cloud structure such as cloud base and top height (CBH, CTH), and cloud thickness are calculated as follows: CBH and CTH are calculated as the first and last pixel within the cloud column. For non-precipitating cloud columns, the first pixel is detected by the ceilometer and the last by the CDR. For precipitating Ice or precipitating Mixed-phase clouds, the first pixel detected by the ceilometer is the Droplets or Drizzle & droplets, which are within the rain layer. These pixels are probably rain pixels, that's why we remove them below the melting layer rather than using the CBH product directly from Cloudnet. Lastly, cloud thickness was calculated as the difference between cloud base and top height pixels.

Multi-layer clouds are problematic due to inaccurate LWP partitioning among cloud layers (Shupe et al., 2015). The partitioning needs an accurate cloud profile, which cannot be assessed with only radar reflectivity. The reflectivity is not sensible to small cloud droplets and the lidar signal is totally attenuated by the first layer. Hence, liquid layers will be poorly identified causing an inaccurate determination of liquid and mixed phase clouds (Nomokonova et al., 2019). Therefore, the following analyses are focused only on single-layer clouds, since their physical properties can be accurately retrieved.

A comparison between CBA and the profile-based algorithm (PBA) (see Appendix A for algorithm description), used by previous studies, is conducted in order to evaluate its applicability. Figure 4 presents a comparison between PBA and CBA applied for a mixed phase on 6 May 2023 in Granada. The cloud phase can be checked in Figure 4.a, which shows a clear layer of mixed phase cloud. Below that, Figure 4.b shows that the PBA fragments the cloud structure into different types. It is easy to observe that the same cloud is being classified as ice or mixed-phase cloud multiple times in the PBA. On the other hand, Figure 4.c shows that the CBA preserves the homogeneity of the cloud and classifies it correctly during the entire cloudy time.

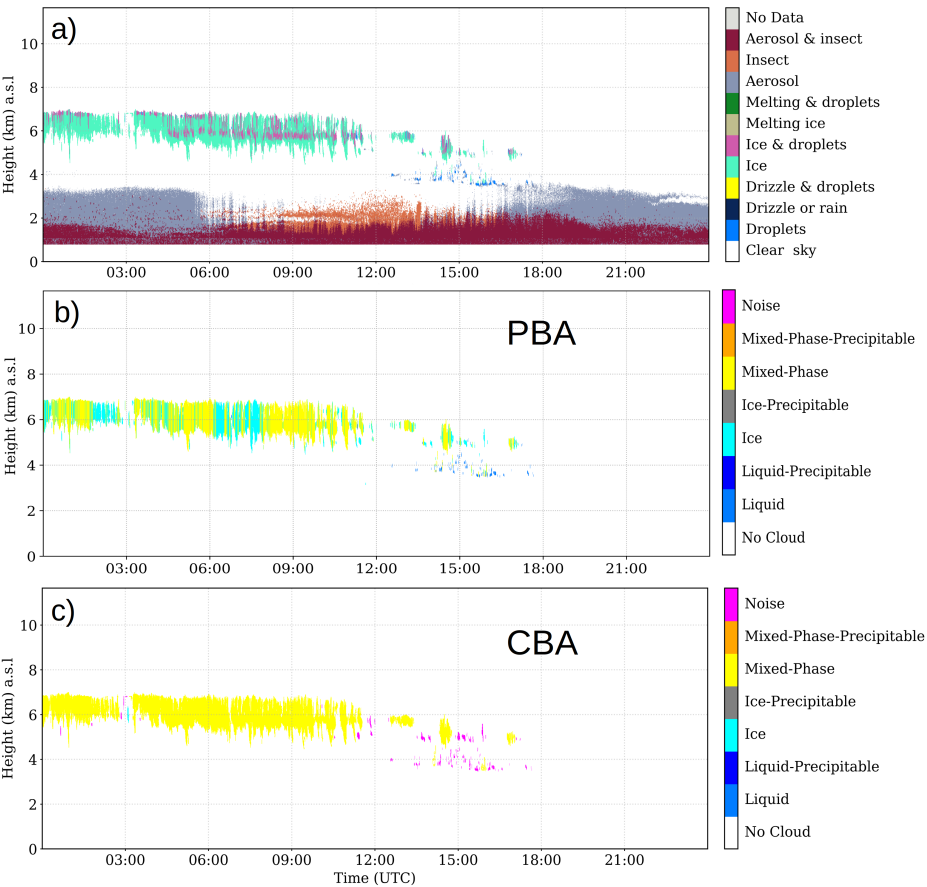


Figure 4. Comparison of cloud typing from different cloud classification algorithms at the AGORA station, on 6 May 2023; (a) HC from Cloudnet, indicating a clear single layer of Mixed-Phase cloud (b) PBA cloud typing, showing an inhomogeneous cloud classification. (c) CBA cloud typing, showing an homogeneous cloud classification.

The inhomogeneous cloud classification of the PBA may result in assigning similar physical properties to different cloud types. As can be clearly seen in Figure 4.b, both ice and mixed phase clouds will have similar daily cloud occurrence and

average CBH. However, this is a consequence of the classification algorithm, which could affect further analysis of cloud type
 180 properties, reaching misleading conclusions.

To assess the impact of the cloud classification algorithm on cloud property statistics, Pearson correlation coefficients between cloud types were calculated based on their daily occurrence. Figure 3.a and Figure 3.b show only the relevant correlations we found for PBA and CBA, respectively. They are correlations between Mixed-Phase and Ice clouds, Liquid and Precipitating Liquid clouds, and Precipitating Mixed-Phase and Precipitating-Liquid clouds.

185 For the PBA, the correlations are 0.39 between ice and mixed-phase clouds and 0.32 between liquid and precipitating liquid. On the other hand, the corresponding correlations for the CBA are 0.09 and 0.19, respectively. As shown, the CBA exhibits weaker or negligible correlations, whereas those from the PBA are more pronounced. The pronounced correlations from PBA are related to its classification scheme, which may link clouds between each other. However, the CBA provides a time-homogenized cloud classification, avoiding the cloud linking. These findings underscore how these two algorithms
 190 associate different cloud types, which can potentially impact cloud property statistics.

Table 3. Comparison of Pearson correlation coefficients for daily cloud occurrence (D.O.) and IWC between different cloud types using CBA and PBA.

Cloud type		Pearson correlation on D.O	
		PBA	CBA
Mixed-Phase	Ice	0.39	0.09
Liquid	Precipitating Liquid	0.32	0.19
Precipitating Liquid	Precipitating Mixed-Phase	0.26	0.04
		Pearson correlation on IWC	
		PBA	CBA
Mixed-Phase	Ice	0.70	0.24

Table 3 shows the correlation of daily average IWP between ice and mixed-phase clouds, for PBA and CBA, respectively. Their corresponding Pearson correlation coefficient (R) are 0.7 and 0.24, almost 3 times higher for PBA. Since there is no evidence that the IWP should correlate between Ice and Mixed-Phase clouds, there is no explanation for such high correlations observed for the PBA. Therefore, this algorithm can generate not explained relations between clouds. In this light, the new
 195 approach shows a promising classification method and will be employed in this study.

4 Results and discussion

4.1 Annual Cloud Variability

Figure 5.a presents the monthly occurrence frequency at AGORA station, for different sky conditions, i.e. single-layer clouds, multi-layer clouds, clear sky, and noise (as defined in section 3.1). The data reveal a distinct seasonal pattern in cloud cover

200 and clear sky occurrences. Clear sky conditions (green line) are predominant most of the year, being most frequent in the summer, peaking in July and August ($> 80\%$), with a significant decline in spring, reaching the lowest point in March and April (40%), when cloudy skies are predominant. This pattern indicates a higher prevalence of clear skies during the warmer months and increased cloudiness during the colder months. Single-layer (blue line) and multi-layer clouds (orange line) exhibit very similar values with a relatively weak seasonal pattern throughout the year showing maxima in spring and minima in the summer months. Multi-layer clouds present a slightly more pronounced seasonal variation than single-layer clouds, with higher occurrence in spring (35% for multi-layer and 25% for single-layer clouds in April) and lower occurrence in summer (3% for multi-layer versus 9% for single-layer clouds in July). The noise category (red line) presents the minimum values and maintains a weak seasonal pattern throughout the year.

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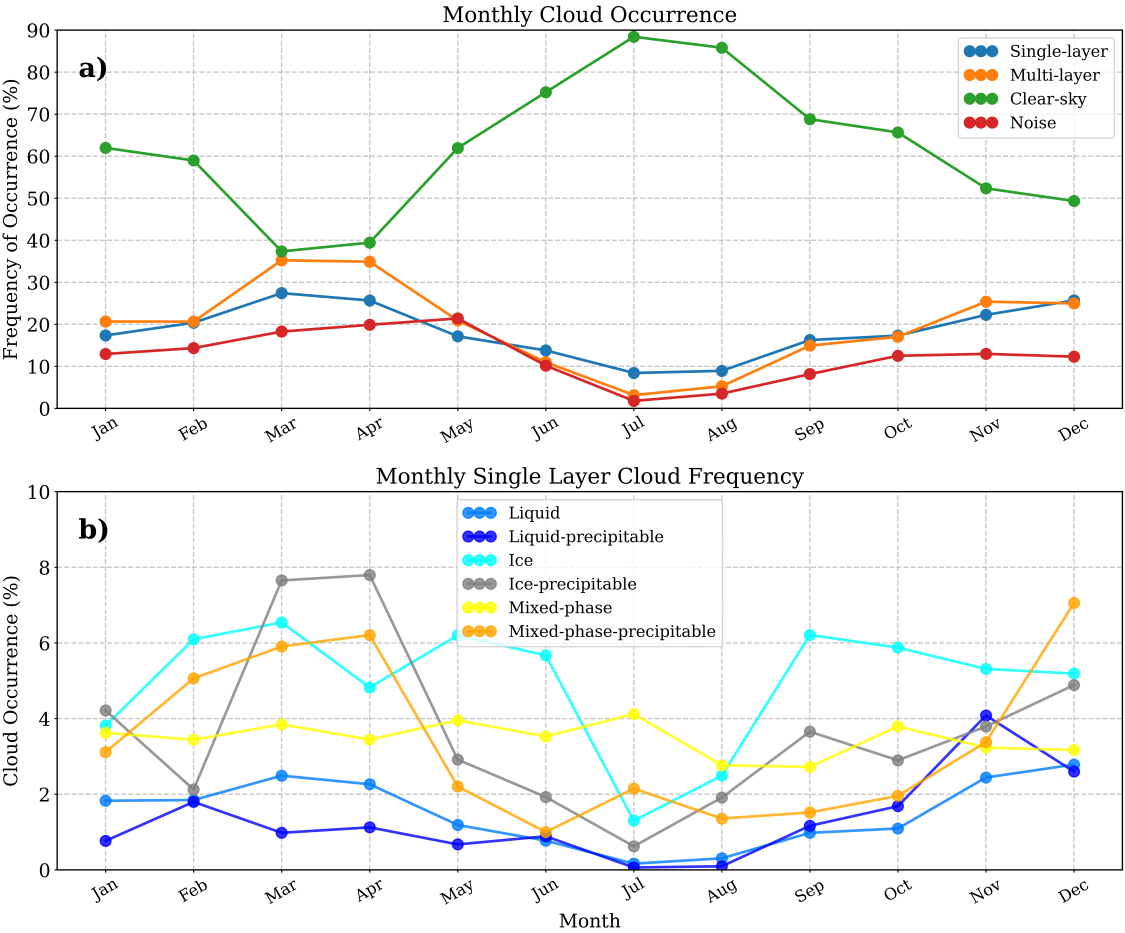


Figure 5. a) Inter-annual frequency of occurrence for single layer clouds (blue), multi-layer clouds (orange), clear sky (green) and noise (red). b)inter-annual single layer cloud occurrence for different cloud categories (see section 3.1). Each cloud types is indicated in the legend

As aforementioned in section 3.1, only single-layer clouds will be analyzed. Thus, the monthly occurrence of single layer
210 clouds per cloud type is depicted in Figure 5.b. The percentage of occurrence in this figure is calculated in relation to the total
number of data. In other words, the sum of cloud occurrences per month in Figure 5.b is the blue curve in Figure 5.a, which is
the single layer frequency in relation to the total number of data.

Liquid clouds (blue) have quite small occurrence, with less than 3% in March and Nov-Dec, when its frequency is higher.
In summer, this cloud has almost zero occurrence (July-Aug), marking an ordinary seasonal trend. Precipitating liquid clouds
215 have less than 2% of occurrence throughout the year, but rises to 4% in November. Clearly, precipitation from liquid clouds
is maxima during Fall. This is the only season where these clouds surpass the occurrence of liquid clouds; however, their
seasonality is ordinary.

Ice clouds (cyan) occurrence is varying around 5% and 7% throughout the year, except in July, when values sharply drop
below 2%. It marks a clear seasonal pattern, with a small presence of ice clouds in summer. Precipitating Ice clouds (gray)
220 have maximum frequency during spring (8%) and winter (5%). Its frequency of occurrence sharply decreases in May, reaching
their minimum in July, with less than 1% of occurrence. This reveals a strong seasonality for liquid precipitation from ice
clouds. Their maximum occurrence is remarkably in spring, decreasing just-after, with lower occurrence than ice clouds for
the remainder period.

Mixed-phase clouds (yellow) exhibit a consistent occurrence around 3% and 4% throughout the year. Its steady frequency
225 indicates an absence of seasonal pattern. Precipitating Mixed-phase clouds (orange) have peaks in April (6%) and December
(7%). Their minimum is also reached in summer, being less than 2% in June and August. This indicates a pronounced season-
ality for liquid precipitation from mixed-phase clouds. Their occurrence exceeds those of Mixed-phase clouds only in spring
and winter.

At the AGORA station, cloud frequency reaches a minimum during summer, when atmospheric stability is pronounced
230 (Bedoya-Velásquez et al., 2019). Particularly, precipitating clouds are more affected due to their strong dependence on atmo-
spheric instability. That's why the strongest seasonal patterns were observed for these clouds. Ice clouds are the only non-
precipitating cloud with a pronounced seasonal pattern. In addition, precipitating clouds with strong seasonal variation were
those with considerable amount of ice, i.e. Precipitating Ice and Precipitating Mixed-phase clouds. Consequently, precipitation
in Granada is highly correlated to these clouds, which are more frequent in Spring. On the other hand, precipitation from Liquid
235 clouds is more frequent in November (Fall).

Despite the high atmospheric stability in summer, a few cloud instances are still being observed. During this season, An-
dalusia experiences easterly winds that transport moisture from the Mediterranean Sea (Bello-Millán et al., 2024). If air masses
are lifted enough to overcome the convective inhibition energy (CINE), conditions become favorable for cloud formation with
substantial vertical development. Since temperatures are high in summer, air masses should be lifted to altitudes higher than
240 other seasons.

The analysis of the diurnal cycles revealed no clear diurnal patterns, thus no analysis on diurnal cycle is shown.

4.2 Seasonal analysis of cloud macrophysics properties

Cloud macrophysics are structure properties such as CBH, CTH and cloud thickness. Their seasonal evolution is displayed in Figure 6. Each color represents a cloud type, where liquid clouds are in blue, ice clouds are in gray and mixed-phase clouds are in orange. Also, the lightest colors represent non precipitating clouds, and the darkest colors represent precipitation clouds (see legend on top of the figure). The values in parenthesis are the number of measurements of non-precipitating clouds (On the left) and precipitating clouds (On the right). Table 4 resumes the median values of CBH and cloud thickness observed in this figure.

Liquid clouds generally exhibit very low CBH, except in summer, which shows an atypical distribution with fewer data points (4,615) than other seasons. Unusually, this season presented high CBH values, with a median of 2.09 km and few cases above the 75th percentile (~ 3.5 km). On the other hand, other seasons presented a 75th percentile of CBH around 2 km, and median CBH varying from 1.31 (winter) to 1.49 (spring) km, as can be seen in Table 4. The CTH exhibits the same seasonal variability as CBH but slightly higher. Thus, a narrow distribution of cloud thickness (Fig. 6.c – zoomed for better visualization) is observed. Precipitating liquid clouds have median CBHs below 1 km, except in summer, with 1.22 km (see Tab. 4). They exhibited a narrower distribution of CBH and CTH compared to liquid clouds. However, these clouds demonstrated higher cloud thickness, which is twice the width of that observed in liquid clouds during Summer-Winter, as shown in Table 4.

Ice clouds show very high CBH, with the 25th percentile above 6 km. Their median CBH is spanning from 7.24 (Winter) to 8.14 km (Fall) (see Tab. 4). The 25/50/75th percentiles of these clouds remain relatively stable across seasons, indicating a well-defined structure that does not exhibit significant seasonal variation. The same pattern was observed for their CTH, which the 75th percentile exceeds 9 km (Fig. 6.e), regardless of the station. Figure 6.f shows that there are Ice clouds with more than 3 km of width. However their median widths fluctuate around 1 km, with maximum in spring (1.27 km), as can be seen in Table 4. Precipitating Ice clouds have low CBHs, with the 25th percentile below 2 km, except in summer when it is around 3 km. The median CBH in winter and spring is consistent at 1.6 km. In contrast, during summer and fall, the median CBH increases to 3.46 km and 2.53 km, respectively. This variation indicates a pronounced seasonality. Figure 6.f shows that these clouds have a broader distribution of Cloud thickness. The median values range from 3 to 4 km, with the lowest in winter (3.07 km) and the highest in summer (3.96 km). This indicates that these clouds are quite thick and exhibit a weak seasonal variability.

Mixed-phase clouds have CBHs at mid-level altitudes, with distributions that vary significantly across seasons (Fig. 6.g). Seasonal variability is more pronounced in the 25th percentile, while the median CBH exhibits moderate variation. The lowest CBH occurs in winter (4.77 km), spring and summer have similar values (5.31 and 5.49 km, respectively), and fall presents the highest values at 6.50 km. Although median values are high in fall, only summer has no CBH below 4 km. Also, summer presented a significant narrow distribution of CBH. The same pattern was observed for CTH, showing a strong seasonal pattern in terms of CBH and CTH. However, median cloud thickness remains relatively stable around 0.67 km, except in summer, slightly rising to 0.78 km. Precipitating Mixed-phase clouds have low CBHs, with median below 2 km, except in summer, when it is 4.47 km. This station showed minimal cloud presence below 2 km, with the 25th percentile exceeding 3 km. Differently, other seasons have clouds nearer to the ground, with median CBH values between 1 and 2 km (see Tab. 4). The same pattern

was observed for CTH, which showed high values in summer. It shows a strong seasonality for CBH and CTH. In terms of cloud thickness, values are ~ 0.5 km thicker than other seasons (see Tab. 4), also showing a strong seasonality in function of this variable. Precipitating mixed-phase clouds have a strong seasonal pattern in terms of CBH and CTH. In general, they have more They presented a strong seasonal pattern in terms C

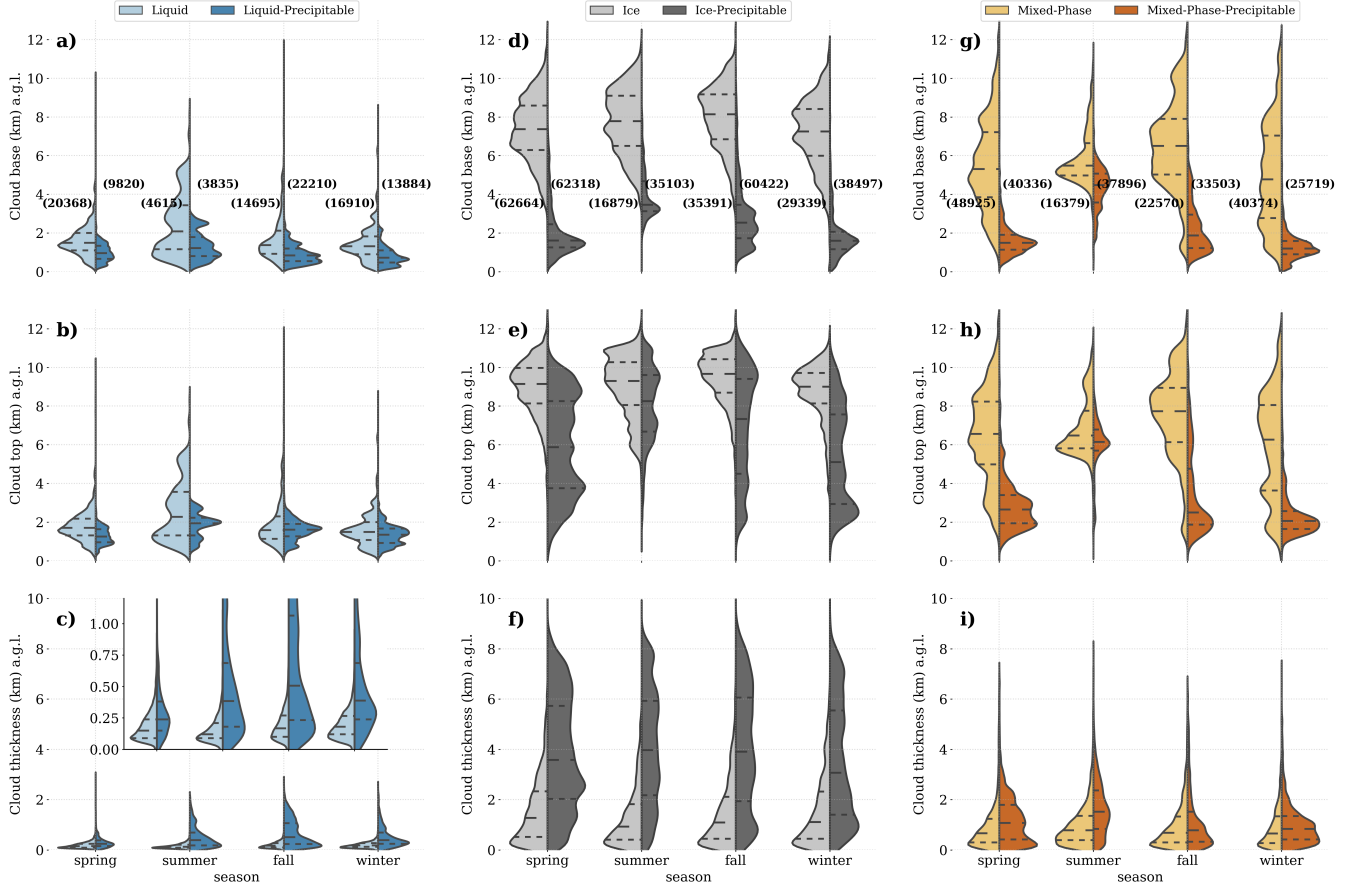


Figure 6. Seasonal evaluation of cloud base height (top), top height (middle) and thickness (bottom). For different cloud types: liquid (left), ice (middle) and mixed-phase (right). Dashed lines indicates 25/50/75th percentiles.

Table 4 shows that Ice-precipitable clouds exhibit the largest cloud thickness, followed by Ice, mixed-phase-precipitable, Mixed-phase, Liquid-precipitable, and Liquid clouds. Precipitating clouds exhibited much larger width than their corresponding non-precipitation cloud. Ice clouds present the highest CBH, followed by Mixed-phase, with liquid-precipitable clouds having the lowest base altitude. Generally, precipitating clouds have much lower CBH than non-precipitating clouds of the same type, and show a stronger seasonal pattern, with higher levels in summer. It is important to note that, despite Cloudnet post-processed data being corrected for liquid attenuation, cloud tops for precipitating clouds can be underestimated due to high liquid attenuation in some particular cases. In addition, for precipitating liquid clouds, CBH is sometimes determined by

"Drizzle & Droplets" pixels. It is hard to know whether these pixels are falling or not when they are inside a “Drizzle or rain” layer. In such cases, the cloud base could be underestimated. At the moment, recent studies are trying to overcome this problem, which can be more intense on marine sites (Roschke et al., 2024), however, it is rarely observed at the AGORA station.

290 In contrast, this is most common for precipitating ice and mixed-phase clouds, where we do not use these pixels to define the cloud base. Instead, we rely on those within the melting layer, as outlined in the methods section.

Table 4. Cloud macrophysics per season and cloud type. Median values of cloud base height, cloud top height, and cloud thickness are shown.

Cloud Properties	Cloud Type	Seasons			
		Spring	Summer	Fall	Winter
Cloud base height (km)	Liquid	1.49	2.09	1.37	1.31
	Liquid-Precipitable	0.95	1.22	0.83	0.72
	Ice	7.36	7.78	8.14	7.24
	Ice-Precipitable	1.61	3.46	2.53	1.60
	Mixed-Phase	5.31	5.49	6.50	4.77
	Mixed-Phase-Precipitable	1.49	4.47	1.86	1.19
Cloud thickness (km)	Liquid	0.15	0.12	0.17	0.18
	Liquid-Precipitable	0.24	0.38	0.51	0.39
	Ice	1.27	0.92	1.09	1.10
	Ice-Precipitable	3.58	3.96	3.91	3.07
	Mixed-Phase	0.66	0.78	0.68	0.66
	Mixed-Phase-Precipitable	1.07	1.52	0.78	0.83

Ice and mixed-phase clouds are the only clouds that do not have higher CBH in summer. Liquid and precipitating Liquid clouds are thicker in fall and winter, while precipitating Ice, Mixed-phase, and precipitating Mixed-phase clouds are thicker in summer. Lastly, Ice clouds are thicker in spring, showing a different seasonal pattern in comparison to Mixed-phase clouds.

295 This behavior contrasts with the findings of Pırloagă et al. (2022), which reported a similar pattern for these clouds in Bucharest. The discrepancy may be attributed to the PBA (profile-based algorithm).

4.3 Cloud microphysical properties

In this section we investigate the inter-annual variability of microphysical retrievals for each category of single-layer clouds detected at the AGORA station. Liquid water content (LWC) and droplet effective radius (r_{liq}) were evaluated for Liquid and

300 Mixed-Phase clouds. Ice water content (IWC) and ice effective radius (r_{ice}) were evaluated for Ice and Mixed-Phase clouds. The analysis was performed for the precipitating case and non-precipitating case.

4.3.1 Liquid clouds

For liquid clouds, Figure 7.c shows the median seasonal profile of LWC on the left, the monthly median profiles of LWC on top-right panel, and the monthly boxplots of LWP on the bottom-right panel. As can be seen on the left panel, Liquid clouds
305 generally have LWC below 250 mg m^{-3} . However, 3 peaks with notably higher values are identified in winter, spring and summer. It shows that for winter and summer, a few clouds with CBHs between 3-4 km contain more water than the low-level clouds. Especially in winter, where more than 1000 mg m^{-3} were found just after 3 km. For spring, a time series analysis (not shown) revealed that high LWC at low levels (below 500 m) were found during spring of 2020, 2021 and 2022. In spring, the monthly median profiles of LWC in the top-right panel of Figure 7.a shows quite large LWC at low-levels in March and
310 May. In winter, high LWC values were found in January and February, while summer presented peaks in June and August. The bottom-right panel shows quite high mean LWP (black star) in April and October. Although the median (red line) stays stable, the average is much higher, highlighting a wide variety of liquid clouds these months. From January to April, and August to October, there is a gradual increase in mean LWP, while the median does not change. This trend corresponds with the increased occurrence of liquid clouds from summer to fall, impacted by different mechanisms of cloud formation.

315 Figure 7.c shows the same cloud properties statistics (LWC and LWP) as shown in Figure 7.c, but for Precipitating Liquid clouds. Note that we avoid the rain contribution in LWP integrating the LWC inside the cloud. The left panel shows that spring and summer have the largest median LWC at low-levels. Especially during summer, which presented almost 2000 mg m^{-3} at $\sim 200 \text{ m}$. Spring showed a minor peak even closer to the ground, and a second one at $\sim 400 \text{ m}$. This indicates that some rain droplets are being mistakenly classified as part of the cloud base. As aforementioned, distinguishing between Drizzle &
320 droplets and Drizzle or rain is challenging for this particular cloud, and some of these pixels (Drizzle & droplets) actually represent rain. Thus, large droplets are attributed to the cloud base, increasing the LWC at low-levels, as observed. Above this level, an increasing trend in LWC up to 2 km is observed for summer, fall, and winter, except in spring. Spring and winter show a peak of $\sim 700 \text{ mg m}^{-3}$ around 2.7 and 2.2 km, and spring and fall present LWC over 3 km of height. The monthly profiles on the top-right panel shows that the peak of LWC in summer is due to high values in June ($> 1000 \text{ mg m}^{-3}$). The LWC peaks
325 in spring are primarily found in April, and the peak observed in winter is located in January. In addition, large values of LWC are also observed in October and November (fall), just after 2 km of height. The bottom panel of Figure 7.c shows the monthly boxplots of LWP, where the 25/75th percentiles show larger variability in LWP in January, suggesting a wide variety of liquid clouds during this month. In contrast, November shows low variability while also having the highest number of registered cloudy profiles (blue line). Spring exhibits consistent LWP across its months, with median below 50 g m^{-2} . Summer shows
330 even less LWP, except for June, as previously observed.

For liquid clouds, the left panel of Figure 7.b shows that the seasonal median profiles of r_{liq} have no significant differences. We observed a sharp increase in droplet size within the first 500 meters, followed by a gradual increase up to 3 km, and then fluctuations around $\sim 6 \mu\text{m}$ at higher altitudes. It suggests a consistent droplet size throughout different atmospheric conditions. These findings underscore the stable nature of liquid clouds in terms of r_{liq} . The top-right panel displays monthly
335 median profiles of r_{liq} , showing that droplet sizes mostly ranges between 3 and $8 \mu\text{m}$. Smaller sizes are observed near the

ground, with larger sizes at higher altitudes, particularly in November, where higher values are seen above 2 km. The bottom panel of Figure 7.c shows the monthly distribution of r_{liq} (violin plots), where the median is stable around 5 μm , with a small decrease in July. Although July has the fewest recorded cloudy profiles (blue line), our dataset is robust and indicates that the frequency of these clouds is indeed very low during this month. Consequently, these limited cloud observations can still
340 effectively represent the general microphysics of these clouds

The median seasonal profiles of r_{liq} (see left panel of Fig. 7.d) for Precipitating Liquid clouds shows significant differences. Spring and summer have similar r_{liq} below 1 km, fluctuating around 5 μm . Above this level, droplet radius increases with highest rate in summers up to 2 km, while spring keeps r_{liq} roughly around 5 μm until 3km, followed by sharp increase to 20 μm . Winter has the second largest r_{liq} at the first 2 km ($\sim 8 \mu\text{m}$), disregarding large rain drizzle particles close to the ground.
345 These particles cause sharp peaks near the surface in summer, fall and winter. In addition, r_{liq} in fall are the largest (below 3 km), reaching 15 μm around 1 km. The monthly median profiles of r_{liq} in the top-right panel of Figure 7.d shows that larger sizes in fall occur in November. Larger sizes related to the first peak below 500 m are associated with Drizzle & droplets pixels linked to rain, as previously observed. The monthly distribution of r_{liq} in the bottom-right panel shows larger sizes in November, as already observed. During spring and early summer, the median is maintained around 10 μm , then decreases in
350 summer, and increases again in fall, when the number of recorded cloudy profiles is maximum.

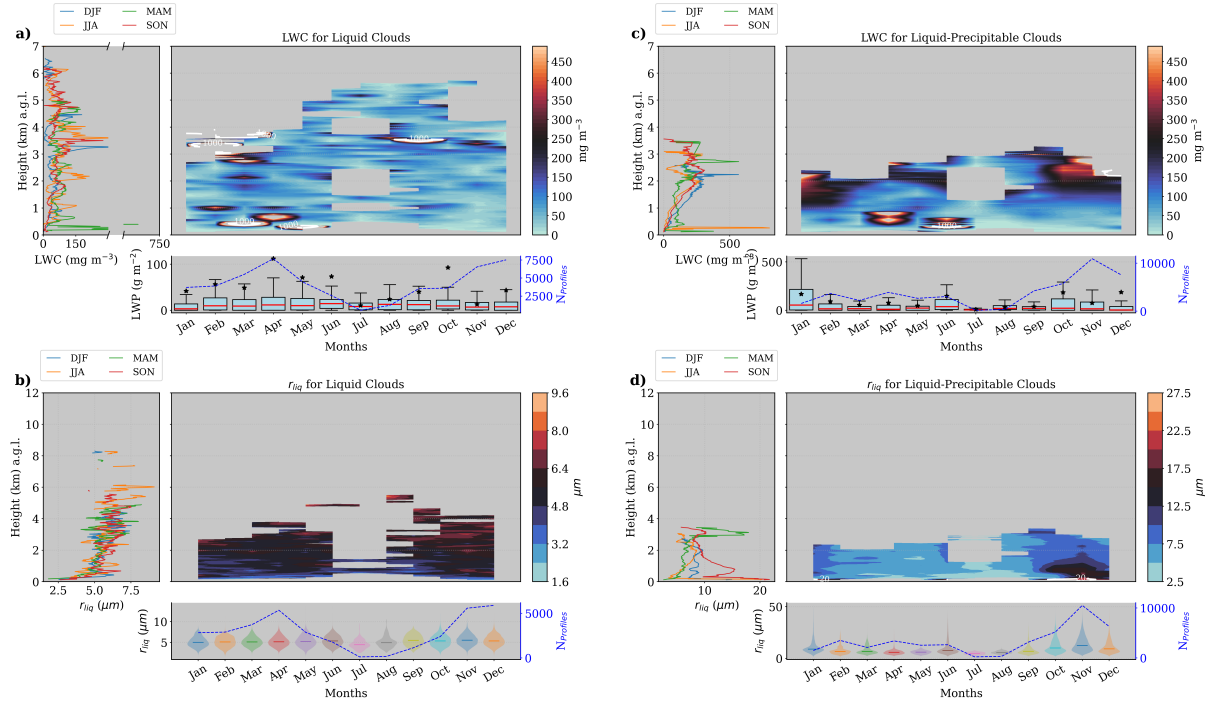


Figure 7. Seasonal and monthly average profiles of microphysical properties, along with monthly of LWP and r_{liq} . Panels a) and b) show LWC, r_{liq} , and LWP for non-precipitating liquid clouds, while panels c) and d) depict the same parameters for precipitating liquid clouds. Specifically: a) and c) display the seasonal (left panel) and monthly (right panel) median profiles of LWC, with boxplots of LWP beneath. Medians are represented by red lines and means by black stars. b) and d) present profiles of r_{liq} for liquid and precipitating liquid clouds, respectively. The left panels show seasonal medians, while the right panels show monthly medians. The monthly distribution of r_{liq} is illustrated using violin plots in the bottom panel.

Liquid clouds generally do not show seasonal patterns in LWC, except in two altitude ranges: 0–1 km and 3–4 km. In the 0–1 km range, a peak of LWC appears in spring, while in the 3–4 km range, LWC peaks in summer, with a slightly lower peak in winter. For r_{liq} , no seasonal pattern was found, and its values are generally around 5 μm . For precipitating liquid clouds, both LWC and r_{liq} are larger than those for Liquid clouds and exhibit a clear seasonal pattern. These properties were influenced by Drizzle & droplets pixels at low levels, as previously discussed, which increases the uncertainty of LWC and r_{liq} near the ground. However, two additional peaks were clearly identified at approximately 2.5 and 2.75 km for winter and spring, respectively. For r_{liq} , larger values were observed in fall up to 3 km, while above this height, spring exhibited the largest sizes, with a pronounced peak around 3 km.

4.3.2 Mixed Phase clouds

360 For Mixed-Phase clouds, Figure 8 presents different panels from the previous figures to account for the presence of both water and ice. Figure 8.a-b display the seasonal and monthly statistics of LWC and r_{liq} , while Figure 8.c-d present the corresponding statistics for IWC and r_{ice} . Precipitating mixed-phase clouds will be analyzed right after.

Figure 8.a shows the median seasonal profile of LWC on the left, the monthly median profiles of LWC on top-right panel, and the monthly boxplots of LWP on the bottom-right panel. As can be seen on the left panel, the median seasonal LWC shows
365 large variability below 4 km and above 10 km. Below 4 km, all seasons exhibit an LWC peak of nearly 200 mg m^{-3} near the ground ($\sim 700 \text{ m}$), except in winter, where this peak is around 400 mg m^{-3} . Then, one more peak was observed in spring (~ 2.8), and two more peaks were observed in fall (~ 2.5 and $3.3 \sim \text{km}$ of height), each exceeding 100 mg m^{-3} . The monthly median profiles of LWC in the top-right panel of Figure 8.a indicate that the second peak in spring occurs in May. The two fall peaks are likely due to October and November, with values around $150\text{--}200 \text{ mg m}^{-3}$ at that altitude. The bottom-right panel
370 of Figure 8.a presents monthly boxplots of LWP, showing that January has the highest mean LWP and the largest difference from the median. This month contributes to the prominent near-ground peak observed in the winter profile. Overall, monthly median LWP (red line) are relatively stable compared to the mean. It indicates a wide variety of Mixed-Phase clouds occurring at different altitudes, as seen in the distinct seasonal peaks.

Figure 8.c shows the same as Figure 8.a, but for IWC. The left panel shows the seasonal median IWC, where it increases
375 from 2 km up to 7 km for winter and spring, and up to 8 km for fall and summer. Winter and fall exhibit nearly identical behavior, with summer following a similar pattern to fall up to nearly 7 km. Above this level, summer has the largest amount of IWC, with a peak of 22 mg m^{-3} around 8 km, and a second peak of 17 mg m^{-3} around 11 km. The monthly median profiles of IWC in the top-right panel of Figure 8.c shows IWC of about $20\text{--}40 \text{ mg m}^{-3}$ between 4 and 10 km, except for July, which presented more elevated IWC above 10 km (more than 60 mg m^{-3}). Furthermore, the monthly boxplots of IWP in the
380 bottom-right panel of Figure 8.c present high differences between the mean and the median during spring, summer and winter. Typically, monthly median IWP are close to zero, being slightly larger in April and summer, when the mean fluctuates between $45\text{--}55 \text{ g m}^{-2}$. It shows the presence of a few Mixed-phase clouds in these months with larger content of ice.

The left panel of Figure 7.b shows the median seasonal profiles of r_{liq} for Mixed-Phase clouds. This panel shows differences between fall and summer at 4 - 5 km, where fall presented a valley, while the other stations presented a peak. Except for fall,
385 median r_{liq} increases with height up to 5 km, where there is almost zero LWC (between 4 - 8 km) as already observed (see Fig. 7.a, left). Summer presented the largest droplets between 4 - 5 km, with $\sim 15 \mu\text{m}$ of radius. The top-right panel of Figure 7.b displays monthly median profiles of r_{liq} , which shows that larger droplets are also found at 10 km ($\sim 12\text{--}18 \mu\text{m}$), in July. Moreover, the monthly profiles show that summer has larger droplets compared to winter and spring, with an effective radius between $14\text{--}22 \mu\text{m}$ at medium heights ($> 4 \text{ km}$ and $< 6 \text{ km}$), and between $12\text{--}18 \mu\text{m}$ around 10 km. The bottom-right panel
390 of Figure 7.b presents the monthly distribution of r_{liq} using violin plots. This panel indicates that droplet sizes are generally around $10 \mu\text{m}$, with distribution modes at even smaller values. In fact, cloud droplets are slightly larger in summer, which is in agreement with the monthly median profiles

Figure 8.c shows the same as Figure 8.d, but for r_{ice} . The seasonal median profiles in the left panel show a smooth decrease in r_{ice} up to 8 km, after which the rate of decrease accelerates until the sizes reach their minimum. In the entire column, summer has a slightly larger radius, especially at 11 km, reaching 30 μm , whereas other seasons showed smaller sizes. The top-right panel displays the monthly median profiles of r_{ice} shows clearly that droplet radius is higher at low levels, where there are low amounts of IWC. However, July also presented r_{ice} larger than 50 μm around 10 km (see Fig. 8.c). The bottom-right panel presents the monthly distribution of r_{ice} , where the median is fluctuating around 45 μm , being slightly larger in summer. Moreover, the mode is above this median, indicating high probability to find even more bigger ice particles.

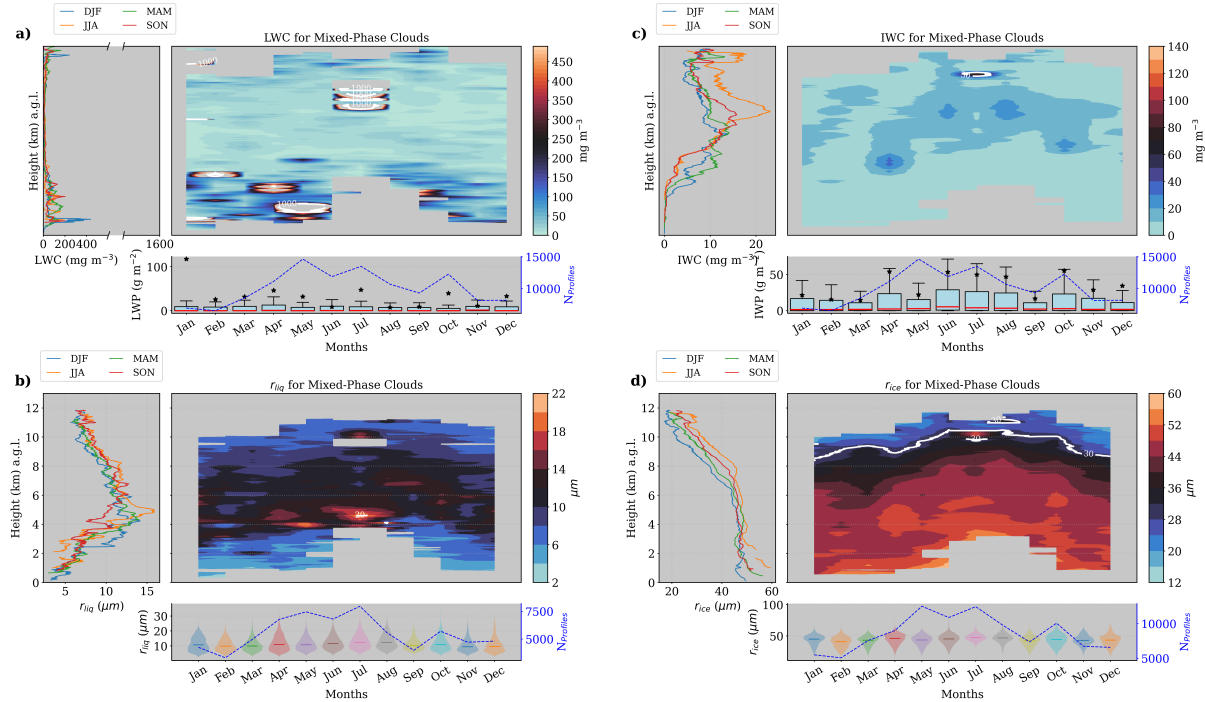


Figure 8. Same as Fig. 7, but for mixed-phase clouds. Seasonal and monthly average of ice water content (IWC) and ice water path (IWP) for mixed-phase clouds. Left panel shows the seasonal median profiles of IWC. Right panel shows the monthly median profiles of IWC and monthly box-plot of IWP in the bottom. For the boxplots, medians are denoted by red lines and the average by black stars.

Figure 9 shows the same as Figure 8, but for Precipitating Mixed-Phase clouds. The left panel of Figure 9.a presents the median seasonal profile of LWC. Fall, winter, and summer display similar profiles up to 2 km, with median LWC values around 100 mg m^{-3} . In contrast, spring shows a higher median LWC, around 200 mg m^{-3} . Above 2 km, winter and spring share similar profiles, while summer and fall also exhibit comparable profiles. Between 2 and 4.5 km it is easy to see that summer and fall have more LWC than winter and spring. The monthly median profiles of LWC in the top-right panel of Figure 9.a indicate that higher LWC values in spring are observed in March and April. Additionally, summer displays larger LWC above 2 km, particularly in July and August. For winter and spring, LWC is primarily located below 2 km, while summer and

fall have more at higher altitudes. This is in agreement with the distribution of CBH. The bottom-right panel of Figure 9.a displays monthly boxplots of LWP, indicating high variability in winter and spring. In these seasons, median values are below 50 g m^{-2} , while mean values exceed 100 g m^{-2} . On the other hand, the few clouds observed in summer exhibit similar LWC with lower variability and reduced LWP

Figure 9.c shows the same as Figure 9.a, but for IWC. The seasonal median profiles on the left-top panel shows a clear seasonal pattern, where IWC is found above 3 km. Typically, summer and spring have similar profiles of median IWC up to an altitude of 9 km. Above this level, summer exhibits a pronounced peak, exceeding 200 mg m^{-3} . Below this level, spring and winter have similar profiles with less IWC. From 3 to 4 km, the median IWC increases up to $22\text{-}33 \text{ mg m}^{-3}$ in spring and winter. Then, values fluctuate around $15\text{-}20 \text{ mg m}^{-3}$ from 4.5 to 8 km. On the other hand, summer and fall presented a maximum of 40 mg m^{-3} at 7 km and 50 mg m^{-3} at 5 km, respectively. The monthly median profiles of IWC in the top-right panel shows that the second peak in summer is due to larger values around June. In fall, the highest values of IWC occur in November, just above 5 km, and ranging from 60 to 100 mg m^{-3} . For winter, the maximum IWC occurs at 5 km in January, reaching more than 120 mg m^{-3} . The bottom-right panel displays monthly boxplots of IWP, indicating that July, August, and September have comparable amounts of IWP, with notably higher values than those observed during the rest of the year. These months have the 75th percentile of IWP around 200 g m^{-2} , while the rest of the year have less than 50 g m^{-2} .

Figure 9.b shows the same as Figure 9.a, but for r_{liq} . The left panel of this figure shows the seasonal median profiles of r_{liq} , where only winter presented larger droplets ($> 15 \mu\text{m}$) near the ground. Other months exhibit a similar growth rate with height up to 4 km, with some deviations. Then, between 4 - 6 km, fall and summer presented the largest droplets. Their respective median r_{liq} is $18 \mu\text{m}$ and $20 \mu\text{m}$, reached around 4.3 km. Spring also has the maximum within the layer (4 - 6 km), but its radius is $\sim 16 \mu\text{m}$, comparable to the winter maximum, at 3.8 km. The top-right panel of Figure 9.b shows the monthly median profiles of r_{liq} . Between 4 - 6 km, this panel shows larger droplets in July, August and September. For summer, larger values of r_{liq} can be clearly identified around 4.5 km in July ($20\text{-}25 \mu\text{m}$) and at 4 km in August ($20 \mu\text{m}$). For spring, larger droplets are found around 7 km ($20\text{-}30 \mu\text{m}$). Lastly, winter shows much smaller droplet sizes, except in January, where larger values are observed near the ground and at 5 km. The bottom-right panel of Figure 9.b displays monthly distribution of r_{liq} . This panel shows completely different shapes among monthly distributions. However, the monthly distribution generally has the median around $10 \mu\text{m}$, reaching $15 \mu\text{m}$ in July, August and September.

The left panel of Figure 9.d shows the seasonal median profiles of r_{ice} . The profiles are similar for summer and fall, as well as for spring and winter, showing a clear seasonal pattern. Summer and fall exhibit median r_{ice} values around $50 \mu\text{m}$ up to 8 km. Above this altitude, r_{ice} sharply decreases in fall, while in summer it peaks at 9.5 km, exceeding $60 \mu\text{m}$. Winter and spring also display r_{ice} around $50 \mu\text{m}$ up to 4 km, after which sizes gradually decrease, reaching a minimum of $20 \mu\text{m}$ at 10 km. The monthly median profiles of r_{ice} in the top-right panel show a clear decrease in size with height, which becomes more pronounced at higher altitudes, especially during summer and fall. In summer, the largest values at 9.5 km are observed in June. The bottom-right panel shows the monthly distribution of r_{ice} , with a narrow distribution across all months and a median generally around $50 \mu\text{m}$.

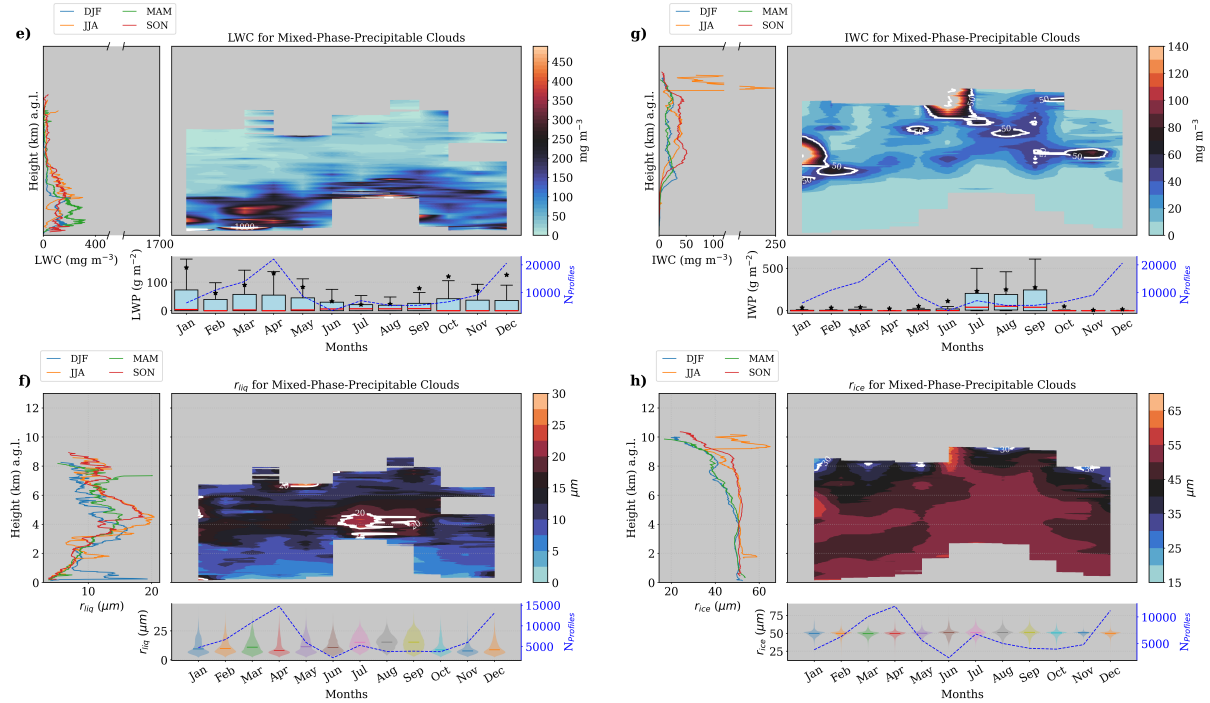


Figure 9. Same as Figure 8, but for precipitating mixed-phase clouds.

Mixed-Phase clouds showed LWC at higher levels than Precipitation Mixed-Phase clouds. For both cloud types, LWC and r_{liq} showed weak seasonal differences. In contrast, IWC and r_{ice} displayed more pronounced seasonal patterns, especially for Precipitation Mixed-Phase clouds, which showed the largest r_{ice} and highest IWC values, along with stronger seasonal variability. These clouds exhibited similar profiles in summer and fall, as well as in winter and spring. Mixed-Phase clouds had higher median r_{liq} at mid-levels, with a gradual increase in droplet size with height up to 6–8 km. On the other hand, r_{ice} generally decreased with height, except for Precipitation Mixed-Phase clouds during summer, where larger particles were observed around 9.5 km, particularly in June.

4.3.3 Ice Clouds

For Ice clouds, Figure 10 shows the same as Figure 7 but for Ice and Precipitating Ice clouds. The top-left panel of Figure 10.a shows the median seasonal profiles of IWC, where elevated values are found near the ground in winter. This indicates a strong seasonal pattern below 3 km, with a pronounced peak of nearly 500 mg m^{-3} in IWC observed at 1 km. Although these clouds exhibit substantially higher IWC near the ground compared to elevated levels, they constitute only a minor fraction of ice clouds, as just 25% have CBH below 6 km. Above 4 km, winter and spring profiles are similar, each with maximum IWC of approximately 25 mg m^{-3} at 6 km. However, spring has an additional peak of similar magnitude around 12 km. Fall also exhibits two peaks, each around 47 mg m^{-3} , at 5 km and 12 km. Lastly, summer shows just one peak of 30 mg m^{-3} around

7 km. The monthly median profiles of IWC are shown in the top-right panel of Figure 10.a, where summer and fall presented largest values of IWC in June and October (exceeding 50 mg m^{-3} between 5 and 6 km). For winter, the extreme values of IWC near the ground are observed in January. This month was influenced by snowfall events, such as the Filomena and Gloria storms. These storms occurred in January of 2020 and 2021, respectively (Buisán et al., 2022; Pérez-González et al., 2022; Marta de Alfonso et al., 2021). The monthly boxplots of IWP in the bottom-right panel of Figure 10.a shows a mean IWP much higher than the median for January, indicating large variability of IWC this month. July shows the smallest values of IWC, in agreement with its minimum cloud occurrence.

Figure 10.c shows the same as Figure 10.a (IWC statistics) but for Precipitating Ice clouds. The top-left panel presents the median seasonal profiles of IWC, showing that summer and fall exhibit similar patterns, while spring and winter also display comparable profiles. For summer and fall, their first maxima coincides at 6 km, with 200 mg m^{-3} . Summer presents a second maxima of about 70 mg m^{-3} around 11 km, and fall has a second maxima overcoming 200 mg m^{-3} just above 12 km. For spring and winter, peaks of 100 and 150 mg m^{-3} are found around 4.5 km, respectively. The monthly median profiles of IWC in the top-right panel shows a clear layer throughout the year, with median IWC values exceeding 50 mg m^{-3} between 2 and 8 km. This layer is higher during summer and fall, with values surpassing 200 mg m^{-3} in June and September. Additionally, July also reaches these values between 9 and 12 km. The monthly boxplots of IWP in the bottom-right panel indicate that these clouds have more IWP during summer and fall. For these periods, the 75th percentile increases from 500 to 1000 g m^{-2} between June and September. In contrast, IWP values remain much lower throughout the rest of the year, particularly in spring and fall, when these clouds are more commonly observed.

Figure 10.b shows the same as Figure 10.a, but for r_{ice} . The top-left panel shows largest sizes below 6 km, especially in winter, where a peak of $56 \mu\text{m}$ at 2 km can be observed. This shows a clear seasonal pattern at low levels, caused by snowfalls in January of 2020 and 2021, as already commented. Also, summer and fall show similar profiles, especially between 6-10 km. Spring and winter also show similar profiles, but from 4-8 km. Above 6 km, the median r_{ice} decreases with height, except in fall, where a pronounced peak is observed around 12 km. The monthly median profiles of r_{ice} in the top-right panel shows that larger sizes at this height are found in October (greater than $30 \mu\text{m}$). The monthly distributions in the bottom panel reveals a stable median at approximately $38 \mu\text{m}$. However, January presents a distribution with a new mode around $50 \mu\text{m}$, showing the presence of larger particles due to snowfall events already discussed.

Figure 10.d shows the same as Figure 9.b, but for Precipitating Ice Clouds. The left panel shows the seasonal median r_{ice} , where summer and fall exhibited similar vertical profiles as well as winter exhibited similar profiles as spring up to 10 km. Since Cloudnet cannot distinguish between ice particles and falling ice particles, larger values close to the ground can be related to graupel. Thus, we will consider profiles variability above 1 km. For winter and spring, r_{ice} increases up to 4.5 km (reaching $50 \mu\text{m}$), whereas in summer and fall, r_{ice} increases up to 6.7 km (reaching $52 \mu\text{m}$). Both peaks have the same magnitude. Above their maximums, particle size decreases up to 10 km, when it increases again for spring, summer and fall. For spring, it increases from $27 \mu\text{m}$ to $33 \mu\text{m}$ at 11 km. For summer, median r_{ice} is kept fluctuating around r_{ice} at $42 \mu\text{m}$ up to 12 km. Lastly, median r_{ice} in fall increases from $21 \mu\text{m}$ at 12 km to $67 \mu\text{m}$ at 12.5 km. The monthly median profiles of r_{ice} in the top-right panel shows that larger particles observed in summer, between 9.5 and 12 km are in July. The second peaks of fall

have not enough neighbor data for painting a contour plot, then they are not shown (but they are still there). Lastly, despite the larger variability with height, the monthly distribution of r_{ice} in the bottom-right panel shows a narrow shape, with the mode and median around $50\ \mu\text{m}$.

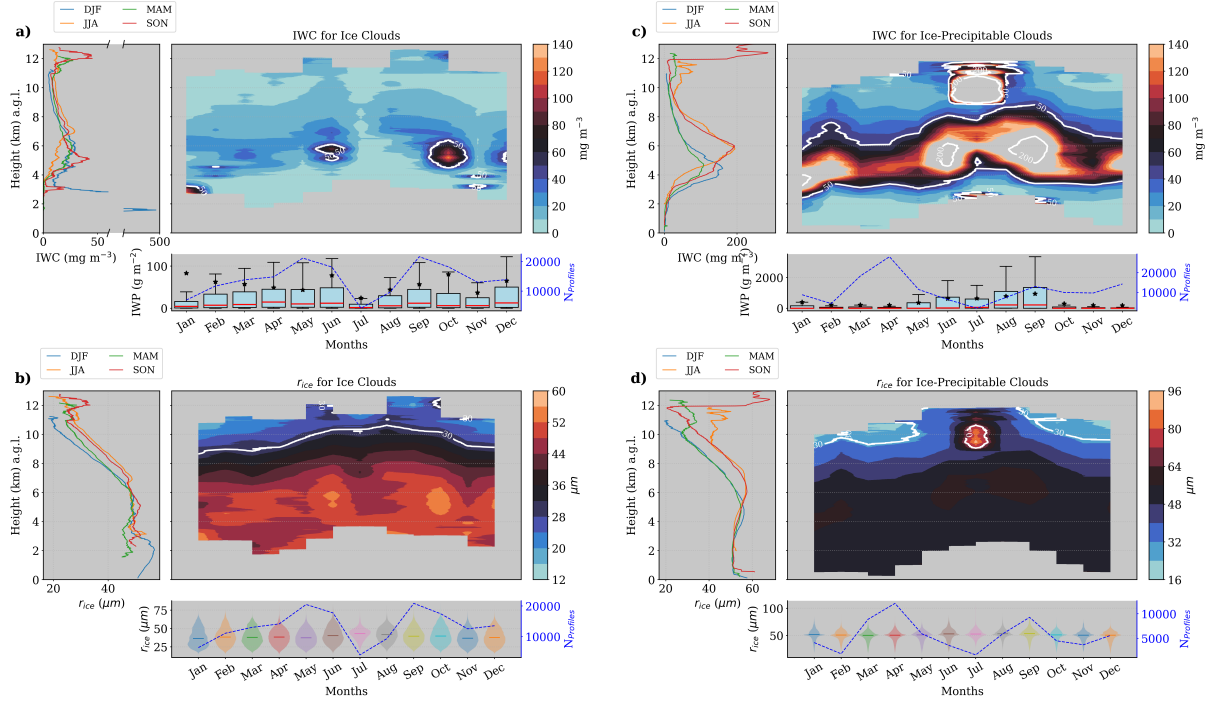


Figure 10. Seasonal and monthly average profiles of microphysical properties, along with monthly IWP and r_{ice} . Panels a) and b) show IWC, r_{ice} , and IWP for non-precipitating ice clouds, while panels c) and d) depict the same parameters for precipitating ice clouds. Specifically: a) and c) display the seasonal (left panel) and monthly (right panel) median profiles of IWC, with boxplots of IWP beneath. Medians are represented by red lines and means by black stars. b) and d) present profiles of r_{ice} for ice and precipitating ice clouds, respectively. The left panels show seasonal medians, while the right panels show monthly medians. The monthly distribution of r_{ice} is illustrated using violin plots in the bottom panel.

Ice clouds exhibited extreme values IWC and r_{ice} near the ground in January. Most of these values were recorded during snowfall events, such as the Filomena and Gloria storms. Generally, these clouds have small particles at higher altitudes, where they are more frequently observed. Also, the IWC shows more seasonality than r_{ice} . However, Precipitation Ice clouds presented a much stronger seasonal pattern, in addition to having more IWC than Ice clouds. Similar to Precipitating Mixed-Phase clouds, we found large amounts of IWC and r_{ice} at high levels in summer ($> 8\ \text{km}$) for Precipitating Ice clouds. Additionally, they demonstrate similar profiles of IWC between summer and spring, as well as spring and fall.

This study presents a statistical analysis of cloud properties using remote sensing ground-based instruments at the AGORA ACTRIS CCRES station, marking the first such analysis in Spain employing the Cloudnet database. Utilizing over half a decade (2018 to 2023) of post-processed data from the Cloudnet consortium, we classified single-layer clouds and retrieved their physical properties using the target classification product. However, this product exhibited some misclassification at our station, necessitating adjustments to avoid misinterpretation of the results. To enhance cloud classification, we compared previous time-based cloud classification methods from recent studies with a novel approach that categorizes clouds into ice (precipitating ice), liquid (precipitating liquid), and mixed-phase (precipitating mixed-phase) clouds. Evaluating this classification method is crucial for understanding the derived cloud properties, as these will be influenced by the chosen method. Consequently, using this new method, we conducted an inter-annual analysis alongside a diurnal cycle analysis of single-layer clouds to assess seasonal variability and identify convective clouds. Additionally, monthly and seasonal analyses of macrophysical and microphysical properties were presented by cloud type to elucidate their differences.

Appendix A: Profile-based algorithm

The Profile-based algorithm classifies clouds by individual profiles (Pîrloagă et al., 2022; Nomokonova et al., 2019; Kneifel et al., 2022). The PBA uses the target classification (TC) from Cloudnet to classify sequences of pixels with height. For each time step, vertical sequences of pixels in TC are classified as follow:

1. Cloud layer criteria: More than 3 consecutive hydrometeor pixels (38-153 m) (Nomokonova et al., 2019) are considered a cloud layer.
 - (a) Liquid criteria: Sequence of only "Droplets" and "Drizzle & Droplets" pixels
 - (b) Ice criteria: Sequence of only "Ice" pixels
 - 520 (c) Mixed-Phase criteria: Sequence of only "Droplets", "Ice", "Ice & Droplets", "Melting ice", "Melting & droplets", "Drizzle and droplets" pixels
 - (d) Rain criteria: For Liquid or Mixed-Phase clouds, if more than 1 pixel of "Drizzle or rain" is found below the cloud layer, then it is considered as a precipitating layer. Thus, this cloud layer is then classified as Precipitating Liquid cloud or Precipitating Mixed-Phase cloud.
- 525 2. Multilayer classification
 - (a) Multi-layer criteria: More than 4 pixels (64-255 m) (Pîrloagă et al., 2022) of any combination of "Clear sky", "Aerosol", "Insect" and "Aerosol & insect" between cloud layers. Otherwise, it's a single layer cloud.

Appendix B: Pixel dilatation

530 The dilation method consists in expanding the ones in the cloud mask to connect the closest groups of hydrometeors, which are likely to belong to the same cloud. The dilatation follows the scheme below, where hydrometeor pixels are represented in blue. Left image represents the original hydrometeor pixels where two groups of clouds were identified. Thus, a dilatation of 1 pixel is applied in all directions, represented by the red pixels in the right image. Therefore, the dilation process will unite these groups with at least 2 pixels of separation distance. Since they are connected, the cloud classification algorithm will save the coordinates of these cloud pixels in just one group (not two, as previously identified).

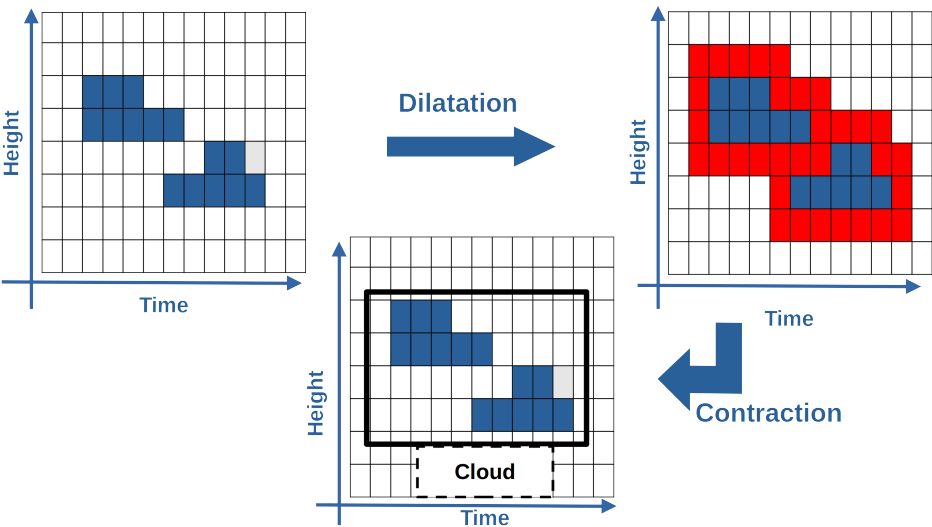


Figure B1. Pixel dilatation and pixel contraction illustration. Left panel shows two hydrometeor groups. Right panel still shows these two clouds and the dilated pixels in red. Since these clouds are connected, the classification algorithm will understand there is only one cloud. Bottom panel shows the result of the pixel contraction, returning cloud pixels to its original position

535 Next, a contraction process is done to return the coordinates of true hydrometeors to the original position. The contraction will remove the fake coordinates of hydrometeor (pixels in red on the above figure). In order to illustrate the method, the left panel of Figure B2 shows the Cloudnet hydrometeor classification product (on 11 February 2021), while the right panel shows all hydrometeors clusters. Groups inside the black dashed lines are considered clusters and will be classified into cloud types. Additionally, clusters that are very close to each other will be merged.

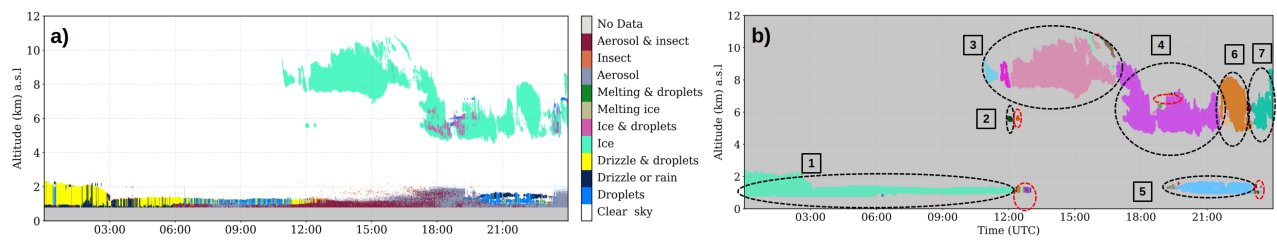


Figure B2. Cloud cluster identification by Cloudnet target classification on 11 February 2021. (a) Target classification, which shows particles typing in time and altitude. (b) Cloud clusters identified in black (dashed lines) and noise (less than 100 pixels connected) by red, dashed lines

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References

- The Earth's Energy Budget, Climate Feedbacks and Climate Sensitivity, in: *Climate Change 2021 – The Physical Science Basis: Working Group I Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, edited by Intergovernmental Panel on Climate Change (IPCC), pp. 923–1054, Cambridge University Press, Cambridge, ISBN 978-1-00-915788-9, <https://doi.org/10.1017/9781009157896.009>, 2023.
- Achtert, P., O'Connor, E. J., Brooks, I. M., Sotiropoulou, G., Shupe, M. D., Pospichal, B., Brooks, B. J., and Tjernström, M.: Properties of Arctic liquid and mixed phase clouds from ship-borne Cloudnet observations during ACSE 2014, <https://doi.org/10.5194/acp-2020-56>, 2020.
- Bedoya-Velásquez, A. E., Navas-Guzmán, F., de Arruda Moreira, G., Román, R., Cazorla, A., Ortiz-Amezcu, P., Benavent-Oltra, J. A., Alados-Arboledas, L., Olmo-Reyes, F. J., Foyo-Moreno, I., Montilla-Rosero, E., Hoyos, C. D., and Guerrero-Rascado, J. L.: Seasonal analysis of the atmosphere during five years by using microwave radiometry over a mid-latitude site, *Atmospheric Research*, 218, 78–89, <https://doi.org/10.1016/j.atmosres.2018.11.014>, 2019.
- Bello-Millán, F. J., Palacios, J., Gutierrez-Castillo, P., and Parras, L.: Simulations of a tornadic supercell event in the south of Spain: Sensitivity to initial and boundary conditions and microphysics parameterizations, *Atmospheric Research*, 300, 107 262, <https://doi.org/10.1016/j.atmosres.2024.107262>, 2024.
- Brenguier, J.-L., Pawlowska, H., and Schüller, L.: Cloud microphysical and radiative properties for parameterization and satellite monitoring of the indirect effect of aerosol on climate, *Journal of Geophysical Research: Atmospheres*, 108, <https://doi.org/10.1029/2002JD002682>, 2003.
- Bühl, J., Seifert, P., Myagkov, A., and Ansmann, A.: Measuring ice- and liquid-water properties in mixed-phase cloud layers at the Leipzig Cloudnet station, *Atmospheric Chemistry and Physics*, 16, 10 609–10 620, <https://doi.org/10.5194/acp-16-10609-2016>, 2016.
- Buisán, S. T., Serrano-Notivol, R., Kochendorfer, J., and Bello-Millán, F. J.: Adjustment of Solid Precipitation during the Filomena Extreme Snowfall Event in Spain: From Observations to “True Precipitation”, <https://doi.org/10.1175/BAMS-D-22-0012.1>, 2022.
- Cazorla, A., Casquero-Vera, J. A., Román, R., Guerrero-Rascado, J. L., Toledano, C., Cachorro, V. E., Orza, J. A. G., Cancillo, M. L., Serrano, A., Titos, G., Pandolfi, M., Alastuey, A., Hanrieder, N., and Alados-Arboledas, L.: Near-real-time processing of a ceilometer network assisted with sun-photometer data: monitoring a dust outbreak over the Iberian Peninsula, *Atmospheric Chemistry and Physics*, 17, 11 861–11 876, <https://doi.org/10.5194/acp-17-11861-2017>, 2017.
- Dougherty, E.: *Mathematical Morphology in Image Processing*, CRC Press, ISBN 978-1-351-83050-8, 2018.
- Frisch, A. S., Graham Feingold, Feingold, G., C. W. Fairall, Fairall, C. W., Taneil Uttal, Uttal, T., J. B. Snider, J. B. Snider, Snider, J. B., and J. B. Snider: On cloud radar and microwave radiometer measurements of stratus cloud liquid water profiles, *Journal of Geophysical Research*, 103, 23 195–23 197, <https://doi.org/10.1029/98jd01827>, 1998.
- Frisch, S., Shupe, M., Djalalova, I., Feingold, G., and Poellot, M.: The Retrieval of Stratus Cloud Droplet Effective Radius with Cloud Radars, 2002.
- Griesche, H. J., Seifert, P., Ansmann, A., Baars, H., Barrientos Velasco, C., Bühl, J., Engelmann, R., Radenz, M., and Zhenping, Y.: Application of the shipborne remote sensing supersite OCEANET for profiling of Arctic aerosols and clouds during Polarstern cruise PS106, <https://doi.org/10.5194/amt-2019-434>, 2019.
- Heese, B., Flentje, H., Althausen, D., Ansmann, A., and Frey, S.: Ceilometer lidar comparison: backscatter coefficient retrieval and signal-to-noise ratio determination, *Atmospheric Measurement Techniques*, 3, 1763–1770, <https://doi.org/10.5194/amt-3-1763-2010>, 2010.

- Hoerling, M., Eischeid, J., Perlwitz, J., Quan, X., Zhang, T., and Pegion, P.: On the Increased Frequency of Mediterranean Drought, <https://doi.org/10.1175/JCLI-D-11-00296.1>, 2012.
- 580 Hogan, R. J. and O'Connor, E. J.: Facilitating cloud radar and lidar algorithms: the Cloudnet Instrument Synergy/Target Categorization product, 2004.
- Hogan, R. J., Bouniol, D., Ladd, D. N., O'Connor, E. J., and Illingworth, A. J.: Absolute Calibration of 94/95-GHz Radars Using Rain, 2003.
- Hogan, R. J., Mittermaier, M. P., and Illingworth, A. J.: The Retrieval of Ice Water Content from Radar Reflectivity Factor and Temperature and Its Use in Evaluating a Mesoscale Model, <https://doi.org/10.1175/JAM2340.1>, 2006.
- 585 Illingworth, A. J., Hogan, R. J., O'Connor, E. J., Bouniol, D., Brooks, M. E., Delanoé, J., Donovan, D. P., Eastment, J. D., Gaussiat, N., Goddard, J. W. F., Haeffelin, M., Baltink, H. K., Krasnov, O. A., Pelon, J., Piriou, J.-M., Protat, A., Russchenberg, H. W. J., Seifert, A., Tompkins, A. M., van Zadelhoff, G.-J., Vinit, F., Willén, U., Wilson, D. R., and Wrench, C. L.: Cloudnet, <https://doi.org/10.1175/BAMS-88-6-883>, 2007.
- Kneifel, S., Pospichal, B., von Terzi, L., Zinner, T., Puh, M., Hagen, M., Mayer, B., Löhnert, U., and Crewell, S.: Multi-year cloud and precipitation statistics observed with remote sensors at the high-altitude Environmental Research Station Schneefernerhaus in the German Alps, *Meteorologische Zeitschrift*, pp. 69–86, <https://doi.org/10.1127/metz/2021/1099>, 2022.
- Küchler, N., Kneifel, S., Löhnert, U., Kollias, P., Czekala, H., and Rose, T.: A W-Band Radar–Radiometer System for Accurate and Continuous Monitoring of Clouds and Precipitation, <https://doi.org/10.1175/JTECH-D-17-0019.1>, 2017.
- Marta de Alfonso, de Alfonso, M., Jue Lin-Ye, Lin-Ye, J., José María García-Valdecasas, García-Valdecasas, J. M., Susana Pérez-Rubio, Pérez-Rubio, S., M. Y. Luna, M. Yolanda Luna, Luna, M. Y., Daniel Santos-Muñoz, Santos-Muñoz, D., María Ángeles Martínez Ruiz, Ruiz, M. I., Begoña Pérez-Gómez, Pérez-Gómez, B., Enrique Álvarez-Fanjul, and Álvarez-Fanjul, E.: Storm Gloria: Sea State Evolution Based on in situ Measurements and Modeled Data and Its Impact on Extreme Values, *Frontiers in Marine Science*, 8, <https://doi.org/10.3389/fmars.2021.646873>, 2021.
- 595 Navas-Guzmán, F., Fernández-Gálvez, J., Granados-Muñoz, M. J., Guerrero-Rascado, J. L., Bravo-Aranda, J. A., and Alados-Arboledas, L.: Tropospheric water vapour and relative humidity profiles from lidar and microwave radiometry, *Atmospheric Measurement Techniques*, 7, 1201–1211, <https://doi.org/10.5194/amt-7-1201-2014>, 2014.
- 600 Nomokonova, T., Ebell, K., Löhnert, U., Maturilli, M., Ritter, C., and O'Connor, E.: Statistics on clouds and their relation to thermodynamic conditions at Ny-Ålesund using ground-based sensor synergy, *Atmospheric Chemistry and Physics*, 19, 4105–4126, <https://doi.org/10.5194/acp-19-4105-2019>, 2019.
- 605 Pérez-González, M. E., García-Alvarado, J. M., García-Rodríguez, M. P., and Jiménez-Ballesta, R.: Evaluation of the Impact Caused by the Snowfall after Storm Filomena on the Arboreal Masses of Madrid, *Land*, 11, 667, <https://doi.org/10.3390/land11050667>, 2022.
- Pîrloagă, R., Ene, D., Boldeanu, M., Antonescu, B., O'Connor, E. J., and Ștefan, S.: Ground-Based Measurements of Cloud Properties at the Bucharest–Măgurele Cloudnet Station: First Results, *Atmosphere*, 13, 1445, <https://doi.org/10.3390/atmos13091445>, 2022.
- Protat, A., Armstrong, A., Haeffelin, M., Morille, Y., Pelon, J., Delanoë, J., and Bouniol, D.: Impact of conditional sampling and instrumental limitations on the statistics of cloud properties derived from cloud radar and lidar at SIRTa, *Geophysical Research Letters*, 33, <https://doi.org/10.1029/2005GL025340>, 2006.
- 610 Protat, A., Bouniol, D., Delanoë, J., O'Connor, E., May, P. T., Plana-Fattori, A., Hasson, A., Görsdorf, U., and Heymsfield, A. J.: Assessment of Cloudsat Reflectivity Measurements and Ice Cloud Properties Using Ground-Based and Airborne Cloud Radar Observations, <https://doi.org/10.1175/2009JTECHA1246.1>, 2009.

- 615 Protat, A., Delanoë, J., O'Connor, E. J., and L'Ecuyer, T. S.: The Evaluation of CloudSat and CALIPSO Ice Microphysical Products Using Ground-Based Cloud Radar and Lidar Observations, <https://doi.org/10.1175/2009JTECHA1397.1>, 2010.
- Rios-Entenza, A., Soares, P. M. M., Trigo, R. M., Cardoso, R. M., and Miguez-Macho, G.: Moisture recycling in the Iberian Peninsula from a regional climate simulation: Spatiotemporal analysis and impact on the precipitation regime, *Journal of Geophysical Research: Atmospheres*, 119, 5895–5912, <https://doi.org/10.1002/2013JD021274>, 2014.
- 620 Roschke, J., Witthuhn, J., Klingebiel, M., Haarig, M., Foth, A., Kötsche, A., and Kalesse-Los, H.: Discriminating between "Drizzle or rain" and sea salt aerosols in Cloudnet for measurements over the Barbados Cloud Observatory, <https://doi.org/10.5194/egusphere-2024-894>, 2024.
- Rose, T., Crewell, S., Löhnert, U., and Simmer, C.: A network suitable microwave radiometer for operational monitoring of the cloudy atmosphere, *Atmospheric Research*, 75, 183–200, <https://doi.org/10.1016/j.atmosres.2004.12.005>, 2005.
- 625 Saavedra Garfias, P., Kalesse-Los, H., von Albedyll, L., Griesche, H., and Spreen, G.: Asymmetries in cloud microphysical properties ascribed to sea ice leads via water vapour transport in the central Arctic, *Atmospheric Chemistry and Physics*, 23, 14 521–14 546, <https://doi.org/10.5194/acp-23-14521-2023>, 2023.
- Schimmel, W., Kalesse-Los, H., Maahn, M., Vogl, T., Foth, A., Garfias, P. S., and Seifert, P.: Identifying cloud droplets beyond lidar attenuation from vertically pointing cloud radar observations using artificial neural networks, *Atmospheric Measurement Techniques*, 15, 5343–5366, <https://doi.org/10.5194/amt-15-5343-2022>, 2022.
- 630 Shupe, M. D., Turner, D. D., Zwink, A., Thieman, M. M., Mlawer, E. J., and Shippert, T.: Deriving Arctic Cloud Microphysics at Barrow, Alaska: Algorithms, Results, and Radiative Closure, <https://doi.org/10.1175/JAMC-D-15-0054.1>, 2015.
- Twomey, S.: The Influence of Pollution on the Shortwave Albedo of Clouds, 1977.
- Yoshida, Y. and Asano, S.: Effects of the Vertical Profiles of Cloud Droplets and Ice Particles on the Visible and Near-Infrared Radiative Properties of Mixed-Phase Stratocumulus Clouds, *Journal of the Meteorological Society of Japan. Ser. II*, 83, 471–480, <https://doi.org/10.2151/jmsj.83.471>, 2005.
- 635 Zeng, X., Gong, J., Li, X., and Wu, D. L.: Modeling the Radiative Effect on Microphysics in Cirrus Clouds Against Satellite Observations, *Journal of Geophysical Research: Atmospheres*, 126, e2020JD033 923, <https://doi.org/10.1029/2020JD033923>, 2021.